

0. Why do you think we assume $\nu > 1$?

We assume $\nu > 1$ because it guarantees that spending time working has desirable effects on the agent's utility. Specifically, note that

$$\begin{aligned}\frac{\partial}{\partial N_t} \left(\frac{N_t^\nu}{\nu} \right) &= N_t^{\nu-1} > 0 \\ \frac{\partial^2}{\partial N_t^2} \left(\frac{N_t^\nu}{\nu} \right) &= (\nu - 1) N_t^{\nu-2} > 0\end{aligned}$$

when $\nu > 1$. This implies that the marginal disutility of working is increasing, in other words, the disutility function of working is convex. This assumption attempts to replicate what we observe in the real world: working an additional hour increases disutility more than the previous additional hour.

1. Write the Lagrangian of the Planner's Problem.

The Lagrangian of the social planner's problem is given by

$$\mathcal{L} = E_t \sum_{j=0}^{\infty} \beta^j \left\{ \ln \left(C_{t+j} - \tau \frac{N_{t+j}^\nu}{\nu} \right) + \lambda_{t+j} \left[Z_{t+j} K_{t+j}^\alpha N_{t+j}^{1-\alpha} + K_{t+j} (1 - \delta) - C_{t+j} - K_{t+j+1} \right] \right\}.$$

$t=0$

2. Calculate the first order conditions.

The derivatives of the Lagrangian are given by:

$$\begin{aligned}C_t : \frac{1}{C_t - \tau \frac{N_t^\nu}{\nu}} &= \lambda_t \\ N_t : \frac{\tau N_t^{\nu-1}}{C_t - \tau \frac{N_t^\nu}{\nu}} &= \lambda_t (1 - \alpha) Z_t \left(\frac{K_t}{N_t} \right)^\alpha \\ K_{t+1} : E_t \left[\beta \lambda_{t+1} \left(\alpha Z_{t+1} \left(\frac{N_{t+1}}{K_{t+1}} \right)^{1-\alpha} + (1 - \delta) \right) - \lambda_t \right] &= 0 \\ \lambda_t : Z_t K_t^\alpha N_t^{1-\alpha} + K_t (1 - \delta) - C_t - K_{t+1} &= 0.\end{aligned}$$

3. Rewrite the system of equations, writing the payments to labor and capital

as a function of the rations Y_t/N_t and Y_t/K_t .

We rewrite the payments to labor and capital using the ratios Y_t/N_t and Y_t/K_t :

$$W_t = (1 - \alpha) Z_t \left(\frac{K_t}{N_t} \right)^\alpha = (1 - \alpha) \frac{Y_t}{N_t}$$

$$R_{t+1} = \alpha Z_t \left(\frac{N_t}{K_t} \right)^{1-\alpha} + (1 - \delta) = \alpha \frac{Y_t}{K_t} + (1 - \delta).$$

Therefore, the first order condition for labor and capital can be rewritten in the following way:

$$\frac{\tau N_t^{v-1}}{C_t - \tau \frac{N_t^v}{v}} = \lambda_t (1 - \alpha) \frac{Y_t}{N_t}$$

$$E_t \left[\beta \lambda_{t+1} \left(\alpha \frac{Y_{t+1}}{K_{t+1}} + (1 - \delta) \right) - \lambda_t \right] = 0$$

Now, we substitute the equations to obtain the expressions we desire. First, note that:

$$\frac{\lambda_{t+1}}{\lambda_t} = \frac{C_t - \tau \frac{N_t^v}{v}}{C_{t+1} - \tau \frac{N_{t+1}^v}{v}},$$

and we use this to simplify the equations:

$$\begin{aligned} \tau N_t^v &= (1 - \alpha) Y_t && \text{(Leisure-consumption)} \\ I_t &= K_{t+1} - (1 - \delta) K_t && \text{(Capital accumulation)} \\ Y_t &= Z_t K_t^\alpha N_t^{1-\alpha} && \text{(Technology)} \\ Y_t &= C_t + I_t && \text{(Feasibility)} \\ \frac{1}{C_t - \tau \frac{N_t^v}{v}} &= \beta E_t \left[\frac{1}{C_{t+1} - \tau \frac{N_{t+1}^v}{v}} \left(\alpha \frac{Y_{t+1}}{K_{t+1}} + (1 - \delta) \right) \right] && \text{(Euler)} \\ \ln Z_t &= (1 - \psi) \ln \bar{Z} + \psi \ln Z_{t-1} + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} D(0, \sigma^2) && \text{(TFP)} \end{aligned}$$

4. Evaluate the equations in the steady state. Use the notation X to indicate the steady state value of the variables.

The non-stochastic steady state is given by the following system:

$$\begin{aligned}\tau N^v &= (1 - \alpha) Y \\ I &= \delta K \\ Y &= Z K^\alpha N^{1-\alpha} \\ Y &= C + I \\ \frac{1}{\beta} &= \alpha \frac{Y}{K} + (1 - \delta).\end{aligned}$$

5. Loglinearize the equations of the system. Use lowercase notation to indicate logarithmic deviations from steady state.

Now, we loglinearize the equations around the non-stochastic steady state.

The leisure-consumption equation is given by

$$\begin{aligned}0 &= (1 - \alpha) Y e^{y_t} - \tau (N e^{n_t})^v \\ 0 &\approx (1 - \alpha) Y y_t - \tau v N^v n_t.\end{aligned}$$

The capital accumulation equation is given by

$$\begin{aligned}0 &= K e^{k_{t+1}} - (1 - \delta) K e^{k_t} - I e^{i_t} \\ 0 &\approx K k_{t+1} - (1 - \delta) K k_t - \delta K i_t \\ 0 &\approx k_{t+1} - (1 - \delta) k_t - \delta i_t.\end{aligned}$$

The technology function is given by

$$\begin{aligned}0 &= Y e^{y_t} - Z e^{z_t} (K e^{k_t})^\alpha (N e^{n_t})^{1-\alpha} \\ 0 &\approx Y y_t - Y z_t - \alpha Y k_t - (1 - \alpha) Y n_t \\ 0 &\approx y_t - z_t - \alpha k_t - (1 - \alpha) n_t.\end{aligned}$$

The feasibility condition is given by

$$\begin{aligned}0 &= Y e^{y_t} - C e^{c_t} - I e^{i_t} \\ 0 &\approx Y y_t - C c_t - I i_t \\ 0 &\approx y_t - \frac{C}{Y} c_t - \frac{I}{Y} i_t.\end{aligned}$$

The Euler equation is given by

$$\begin{aligned}
 0 &= \beta E_t \left[\frac{1}{C e^{c_{t+1}} - \tau \frac{(N e^{n_{t+1}})^v}{v}} \left(\alpha \frac{Y e^{y_{t+1}}}{K e^{k_{t+1}}} + (1 - \delta) \right) \right] - \frac{1}{C e^{c_t} - \tau \frac{(N e^{n_t})^v}{v}} \\
 0 &\approx \beta E_t \left[\frac{-1}{\beta} \frac{C c_{t+1}}{\left(C - \tau \frac{N^v}{v} \right)^2} + \frac{1}{\beta} \frac{\tau N^v n_{t+1}}{\left(C - \tau \frac{N^v}{v} \right)^2} + \frac{\alpha \frac{Y}{K}}{\left(C - \tau \frac{N^v}{v} \right)} (y_{t+1} - k_{t+1}) \right] + \frac{C c_t}{\left(C - \tau \frac{N^v}{v} \right)^2} - \frac{\tau N^v n_t}{\left(C - \tau \frac{N^v}{v} \right)^2} \\
 0 &\approx E_t [\tau N^v n_{t+1} - C c_{t+1}] + \alpha \frac{Y}{K} \left(C - \tau \frac{N^v}{v} \right) \beta E_t [y_{t+1} - k_{t+1}] - (\tau N^v n_t - C c_t).
 \end{aligned}$$

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Therefore, the linear stochastic system of equations is given by

$$(1 - \alpha) Y y_t = \tau v N^v n_t \quad (\text{Leisure-consumption})$$

$$k_{t+1} = (1 - \delta) k_t + \delta i_t \quad (\text{Capital accumulation})$$

$$y_t = z_t + \alpha k_t + (1 - \alpha) n_t \quad (\text{Technology})$$

$$y_t = + \frac{C}{Y} c_t + \frac{I}{Y} i_t \quad (1 - \beta(1 - \delta)) \quad (\text{Feasibility})$$

$$\frac{\tau N^v n_t - C c_t}{C - \tau \frac{N^v}{v}} = E_t \left[\frac{\tau N^v n_{t+1} - C c_{t+1}}{C - \tau \frac{N^v}{v}} \right] + \alpha \frac{Y}{K} \beta E_t [y_{t+1} - k_{t+1}] \quad (\text{Euler})$$

$$\begin{aligned}
 & \left(\begin{aligned} z_{t+1} &= \psi z_t + \epsilon_{t+1} & (\text{TFP}) \\ \frac{1}{C - \tau \frac{N^v}{v}} E_t [\tau N^v (n_{t+1} - n_t) + \tau N^v (n_{t+1} - n_t)] &= 1 - \beta(1 - \delta) E_t [y_{t+1} - k_{t+1}] \end{aligned} \right.
 \end{aligned}$$

6. Write a program in Matlab that solves the system. It is mandatory to use the program `exampl1modifv1.m` that is in Canvas as a starting point.

6.1. What equation would you use to calibrate the parameter τ ? Include it in your Matlab program.

To calibrate τ we would use the leisure-consumption equation. We can solve for τ to obtain

$$\tau = (1 - \alpha) \frac{Y}{N^v}.$$

6.2. What do the vectors x_t, y_t, z_t used in the general uhlig notation correspond to in this case?

8

$$\mathbb{E}_t [\tau N^v n_{t+1} - C_{t+1}] + \kappa y / \kappa (C - \tau N^v / v) \beta \mathbb{E}_t [y_{t+1} - k_{t+1}] - (\tau N^v n_t - C_t) = 0$$

$$= \frac{1}{C - \tau N^v / v} \mathbb{E}_t [\tau N^v (n_{t+1} - n_t) - C (C_{t+1} - C_t)]$$

$$+ \underbrace{\alpha^s / \kappa}_{\text{in ss}} \beta \mathbb{E}_t [y_{t+1} - k_{t+1}] = 0$$

$$= \frac{1}{C - \tau N^v / v} \mathbb{E}_t [\tau N^v (n_{t+1} - n_t) - C (C_{t+1} - C_t)]$$

$$+ \underbrace{\beta (1/\beta - (1-\delta))}_{(1-\beta(1-\delta))} \mathbb{E}_t (y_{t+1} - k_{t+1}) = 0$$

$$= \frac{1}{C - \tau N^v / v} \mathbb{E}_t [C (C_{t+1} - C_t) - \tau N^v (n_{t+1} - n_t)] = (1 - \beta(1-\delta)) \mathbb{E}_t (y_{t+1} - k_{t+1})$$

The vectors in Uhlig's notation correspond to the following in our set up:

$$\begin{aligned} x_t &= [k_{t+1}]_{1 \times 1} \\ y_t &= [c_t, y_t, n_t, i_t]_{4 \times 1} \\ z_t &= [z_t]_{1 \times 1}. \end{aligned}$$

6.3. Fill in the matrices $A, B, C, D, F, G, H, J, L, M, N$. Report them.

Using Uhlig's notation, we can write the system in the following way

$$0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t \quad (1)$$

$$0 = E_t [Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t] \quad (2)$$

$$z_{t+1} = Nz_t + \varepsilon_{t+1} \quad (3)$$

$$0 = E_t [\varepsilon_{t+1}]. \quad (4)$$

We rewrite the loglinearized equations in order to be consistent with the notation

$$\begin{aligned} 0 &= (1 - \alpha) Y y_t - \tau v N^v n_t \\ 0 &= k_{t+1} - (1 - \delta) k_t - \delta i_t \\ 0 &= y_t - z_t - \alpha k_t - (1 - \alpha) n_t \\ 0 &= y_t - \frac{C}{Y} c_t - \frac{I}{Y} i_t \\ 0 &= E_t \left[\frac{\tau N^v n_{t+1} - C c_{t+1}}{C - \tau \frac{N^v}{v}} \right] + \alpha \frac{Y}{K} \beta E_t [y_{t+1} - k_{t+1}] - \frac{\tau N^v n_t - C c_t}{C - \tau \frac{N^v}{v}} \\ z_{t+1} &= \psi z_t + \varepsilon_{t+1}. \end{aligned}$$

Then, the values of the matrices are given by

$$\begin{aligned} A &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -(1 - \delta) \\ -\alpha \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & (1 - \alpha) Y & -\tau v N^v & 0 \\ 0 & 0 & 0 & -\delta \\ 0 & 1 & -(1 - \alpha) & 0 \\ -\frac{C}{Y} & 1 & 0 & -\frac{I}{Y} \end{bmatrix} \\ D &= \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \quad F = [0], \quad G = \left[-\alpha \beta \frac{Y}{K} \right], \quad H = [0] \\ &\quad \downarrow \\ &\quad (- (1 - \beta (1 - \delta))) \\ &\quad \downarrow \\ &\quad \beta (1 - \delta) - 1 \end{aligned}$$

$$J = \begin{bmatrix} \frac{-C}{C-\tau\frac{N^v}{v}} & \alpha\beta\frac{Y}{K} & \frac{\tau N^v}{C-\tau\frac{N^v}{v}} & 0 \end{bmatrix}, \quad K = \begin{bmatrix} \frac{C}{C-\tau\frac{N^v}{v}} & 0 & \frac{-\tau N^v}{C-\tau\frac{N^v}{v}} & 0 \end{bmatrix}$$

$$L = [0], \quad M = [0], \quad N = [\psi]$$

6.4. Calculate the solution matrices: P, Q, R, S . Report them.

A solution is given by

$$x_t = Px_{t-1} + Qz_t$$

$$y_t = Rx_{t-1} + Sz_t.$$

Notice equation (1) can be rewritten as $y_t = -C^{-1}(Ax_t + Bx_{t-1} + Dz_t)$, which we plug into equation (2). This gives us

$$0 = E_t \left[Fx_{t+1} + Gx_t + Hx_{t-1} - JC^{-1}(Ax_{t+1} + Bx_t + Dz_{t+1}) \right]$$

$$+ E_t \left[-KC^{-1}(Ax_t + Bx_{t-1} + Dz_t) + Lz_{t+1} + Mz_t \right]$$

$$0 = E_t \left[(F - JC^{-1}A)x_{t+1} + (G - JC^{-1}B - KC^{-1}A)x_t + (H - KC^{-1}B)x_{t-1} \right]$$

$$+ E_t \left[(L - JC^{-1}D)z_{t+1} + (M - KC^{-1}D)z_t \right].$$

Distributing the expectation we obtain

$$0 = \tilde{F}E_tx_{t+1} + \tilde{G}E_tx_t + \tilde{H}E_tx_{t-1} + \tilde{L}E_tz_{t+1} + \tilde{M}E_tz_t$$

$$0 = \tilde{F}E_tx_{t+1} + \tilde{G}x_t + \tilde{H}x_{t-1} + \tilde{L}Nz_t + \tilde{M}z_t$$

$$0 = \tilde{F}E_tx_{t+1} + \tilde{G}x_t + \tilde{H}x_{t-1} + (\tilde{L}N + \tilde{M})z_t.$$

Notice that $E_tx_{t+1} = Px_t + QE_tz_{t+1} = Px_t + QNz_t$, which we substitute into the expression, yielding

$$0 = \tilde{F}Px_t + \tilde{G}x_t + \tilde{H}x_{t-1} + (\tilde{L}N + \tilde{M} + \tilde{F}QN)z_t$$

$$0 = \tilde{F}P(Px_{t-1} + Qz_t) + \tilde{G}(Px_{t-1} + Qz_t) + \tilde{H}x_{t-1} + (\tilde{L}N + \tilde{M} + \tilde{F}QN)z_t$$

$$0 = (\tilde{F}P^2 + \tilde{G}P + \tilde{H})x_{t-1} + (\tilde{L}N + \tilde{M} + \tilde{F}QN + \tilde{F}PQ + \tilde{G}Q)z_t.$$

Since this must be true for all values of x_{t-1}, z_t , then the coefficients must be zero, that is

$$0 = \tilde{F}P^2 + \tilde{G}P + \tilde{H}$$

$$0 = \tilde{L}N + \tilde{M} + \tilde{F}QN + \tilde{F}PQ + \tilde{G}Q,$$

with

$$\begin{aligned}
\tilde{F} &= F - JC^{-1}A \\
\tilde{G} &= G - JC^{-1}B - KC^{-1}A \\
\tilde{H} &= H - KC^{-1}B \\
\tilde{L} &= L - JC^{-1}D \\
\tilde{M} &= M - KC^{-1}D,
\end{aligned}$$

which is a system that allows us to calculate the values of P and Q . Since P is a scalar, we can easily solve the quadratic equation associated to it. The solution is given by

$$P = \frac{-\tilde{G} \pm \sqrt{\tilde{G}^2 - 4\tilde{F}\tilde{H}}}{2\tilde{F}},$$

which we can substitute into the second equation to find the value of Q . Note that we choose the solution for which $P < 1$ to ensure stability.

To calculate R and S we simply substitute the values of P and Q into equation (1) in the following way:

$$\begin{aligned}
y_t &= -C^{-1}(Ax_t + Bx_{t-1} + Dz_t) \\
y_t &= -C^{-1}(A(Px_{t-1} + Qz_t) + Bx_{t-1} + Dz_t) \\
y_t &= -C^{-1}(AP + B)x_{t-1} - C^{-1}(AQ + D)z_t,
\end{aligned}$$

but this must be equal to the proposed optimal policy function for y_t , so this implies that

$$\begin{aligned}
R &= -C^{-1}(AP + B) \\
S &= -C^{-1}(AQ + D).
\end{aligned}$$

6.5. The objective of this question is for you to program the impulse response function. Calculate the response of vector y_t to a 1% shock, as in Uhlig's programs. Suppose that $T = 150$. Graph the vector y_t against time. Make your graph in such a way that at $t = 0$ the first response of the economy to the shock appears. Include in the graph $t = -1$, to see $y = 0$ before the shock. After the shock, graph the periods from $t = 1$ until $t = 33$. It must be part of the program that solves the model.

6.6. Report by hand the first give values of output y_t , in response to the shock.